

UEN006 : TECHNOLOGIES FOR SUSTAINABLE DEVELOPMENT

L	T	P	Cr
2	0	0	2.0

Course Objective: To provide acquaintance with modern cleaner production processes and emerging energy technologies; and to facilitate understanding the need and application of green and renewable technologies for sustainable development of the Industry/society.

Syllabus

Concepts of Sustainability and Industrial Processes: Industrialization and sustainable development; Cleaner production (CP) in achieving sustainability; Source reduction techniques - Raw material substitution; Process modification and equipment optimization; Product design or modification; Reuse and recycling strategies; Resources and by-product recovery from wastes; Treatment and disposal; CDM and Pollution prevention programs; Good housekeeping; CP audits,

Green Design: Green buildings - benefits and challenges; public policies and market-driven initiatives; Effective green specifications; Energy efficient design; Passive solar design; Green power; Green materials and Leadership in Energy and Environmental Design (LEED)

Renewable and Emerging Energy Technologies: Introduction to renewable energy technologies- Solar; wind; tidal; biomass; hydropower; geothermal energy technologies; Emerging concepts; Biomolecules and energy; Fuel cells; Fourth generation energy systems.

Laboratory Work: NA

Course Learning Objectives (CLO)

Upon completion of the course, the students will be able to:

1. comprehend basic concepts in source reduction, waste treatment and management
2. Identify and plan cleaner production flow charts/processes for specific industrial sectors
3. examine and evaluate present and future advancements in emerging and renewable energy technologies

Recommended Books:

1. Kirkwood, R,C, and Longley, A,J, (Eds.), *Clean Technology and the Environment*, Chapman & Hall, London (1995),
2. World Bank Group; *Pollution Prevention and Abatement Handbook – Towards Cleaner Production*, World Bank and UNEP; Washington DC (1998),
3. Modak, P,, Visvanathan, C, and Parasnis, M,, *Cleaner Production Audit, Course Material on Cleaner Production and Waste Minimization*; United Nations Industrial Development Organization (UNIDP) (1995),
4. Rao, S, and Parulekar, B,B,, *Energy Technology: Non-conventional; Renewable and Conventional*; Khanna Pub,(2005) 3rd Ed,